

Reliable Rentals

Database Implementation Report

Part 3 — Physical Implementation

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GitHub Repository: <https://github.com/Elk-dev/DSC623part3>

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1. Introduction

This report documents the physical implementation of the Reliable Rentals database system. Building on the conceptual and logical design phases, this document covers schema creation, seed data population, five operational queries using embedded SQL, and deployment to GitHub with automated execution through GitHub Actions.

1.1 Scope

- Full database schema created in SQLite using embedded SQL in Python
- All five relations populated with seed data (minimum five tuples each)
- Five operational queries matching the user transactions from the logical design report
- Automated execution via GitHub Actions on every push to main
- Live query results published directly to the repository README

1.2 Technology Stack

- Language: Python 3.12
- Database: SQLite via Python sqlite3 standard library
- Version Control: Git and GitHub
- Automation: GitHub Actions

1.3 Repository Structure

The project is organized as follows:

reliable-rentals/		
schema/	create_tables.py	table creation with all constraints
seed/	insert_data.py	seed data for all five relations
queries/	queries.py	five embedded SQL queries
src/	main.py	entry point, calls all modules
.github/	workflows/	GitHub Actions pipeline
README.md		live query results, auto-updated

2. Database Schema

The schema is implemented in `schema/create_tables.py`. All five relations from the logical design are created with their full constraint sets embedded as SQL strings passed to `cursor.executescript()`. Foreign key enforcement is enabled at the connection level before any tables are created.

```
conn.execute("PRAGMA foreign_keys = ON")
```

Tables are created in the following dependency order to satisfy foreign key constraints: OUTLET first, then VEHICLE and STAFF which both reference OUTLET, then CLIENT independently, and finally HIRE_AGREEMENT which references both CLIENT and VEHICLE.

OUTLET

Attribute	Data Type	Key	Constraints	Description
OutletNumber	TEXT	PK	NOT NULL, UNIQUE	Unique outlet identifier
Address	TEXT		NOT NULL	Full street address
PhoneNumber	TEXT		NOT NULL	Contact phone number
FaxNumber	TEXT		NOT NULL, != PhoneNumber	Fax number for the outlet

VEHICLE

Attribute	Data Type	Key	Constraints	Description
RegistrationNumber	TEXT	PK	NOT NULL, UNIQUE	Vehicle registration plate
Model	TEXT		NOT NULL	Vehicle model name
Make	TEXT		NOT NULL	Vehicle manufacturer
EngineSize	TEXT		NOT NULL	Engine displacement
Capacity	INTEGER		NOT NULL, >= 1	Passenger seating capacity
CurrentMileage	INTEGER		NOT NULL, >= 0	Odometer reading
DailyHireRate	REAL		NOT NULL, > 0	Cost per day to hire
OutletNumber	TEXT	FK	NOT NULL	References OUTLET(OutletNumber)

STAFF

Attribute	Data Type	Key	Constraints	Description
StaffNumber	TEXT	PK	NOT NULL, UNIQUE	Unique staff identifier
FirstName	TEXT		NOT NULL	First name
LastName	TEXT		NOT NULL	Last name
HomeAddress	TEXT		NOT NULL	Residential address
HomePhoneNumber	TEXT		NOT NULL	Personal phone number
DateOfBirth	DATE		NOT NULL	Date of birth
Sex	TEXT		NOT NULL, IN (M,F,O)	Sex as recorded
DateJoined	DATE		NOT NULL	Employment start date
JobTitle	TEXT		NOT NULL	Job role or title
Salary	REAL		NOT NULL, > 0	Annual salary
OutletNumber	TEXT	FK	NOT NULL	References OUTLET(OutletNumber)

CLIENT

Attribute	Data Type	Key	Constraints	Description
ClientNumber	TEXT	PK	NOT NULL, UNIQUE	System-generated client ID
FirstName	TEXT		NOT NULL	First name
LastName	TEXT		NOT NULL	Last name
HomeAddress	TEXT		NOT NULL	Residential address
PhoneNumber	TEXT		NOT NULL	Contact phone number
DateOfBirth	DATE		NOT NULL	Date of birth
DrivingLicenseNumber	TEXT	AK	NOT NULL, UNIQUE	Alternate key — government issued

HIRE_AGREEMENT

Attribute	Data Type	Key	Constraints	Description
HireNumber	TEXT	PK	NOT NULL, UNIQUE	Unique hire identifier
StartDate	DATE		NOT NULL, <= EndDate	Date hire commences
EndDate	DATE		NOT NULL, >= StartDate	Planned return date

MileageBefore	INTEGER		NOT NULL, >= 0	Odometer at hire start
MileageAfter	INTEGER		nullable, >= MileageBefore	Odometer at return — NULL if not yet returned
ClientNumber	TEXT	FK	NOT NULL	References CLIENT(ClientNumber)
RegistrationNumber	TEXT	FK	NOT NULL	References VEHICLE(RegistrationNumber)

3. Seed Data

Seed data is implemented in `seed/insert_data.py` using parameterized INSERT OR IGNORE statements to prevent duplicate key errors on repeated runs. Data is inserted in foreign key dependency order. Each relation contains a minimum of five tuples, with the hire agreement data designed specifically to produce meaningful results across all five queries.

3.1 OUTLET — 5 tuples

OutletNumber	Address	PhoneNumber	FaxNumber
O001	12 High Street, Edinburgh, EH1 1TB	0131-555-0101	0131-555-0102
O002	45 Sauchiehall Street, Glasgow, G2 3AH	0141-555-0201	0141-555-0202
O003	8 Union Street, Aberdeen, AB10 1DQ	01224-555-0301	01224-555-0302
O004	22 Princes Street, Dundee, DD1 4ER	01382-555-0401	01382-555-0402
O005	5 Castle Road, Inverness, IV2 3EA	01463-555-0501	01463-555-0502

3.2 VEHICLE — 7 tuples

RegNumber	Model	Make	Engine	Cap	Mileage	Rate/Day	Outlet
EH21 XYZ	Corsa	Vauxhall	1.2L	5	12000	35.00	O001
GW19 ABC	Focus	Ford	1.6L	5	23000	40.00	O002
AB18 DEF	Qashqai	Nissan	1.5L	5	45000	55.00	O003
DD20 GHI	Fiesta	Ford	1.0L	5	8000	30.00	O004
EH22 JKL	Golf	Volkswagen	2.0L	5	31000	50.00	O001
GW21 MNO	Astra	Vauxhall	1.4L	5	17500	38.00	O002
AB20 PQR	Kuga	Ford	2.0L	7	52000	60.00	O003

3.3 STAFF — 6 tuples

StaffNo	FirstName	LastName	DOB	Sex	Joined	JobTitle	Outlet
S001	James	MacDonald	1985-06-15	M	2010-03-01	Branch Manager	O001
S002	Fiona	Campbell	1990-11-22	F	2015-07-14	Sales Advisor	O001
S003	Hamish	Stewart	1978-04-30	M	2008-01-20	Branch Manager	O002
S004	Morag	Fraser	1995-08-10	F	2019-09-03	Sales Advisor	O002
S005	Callum	Ross	1988-02-17	M	2012-05-11	Branch Manager	O003
S006	Isla	Murray	1993-12-05	F	2018-04-22	Fleet Coordinator	O003

3.4 CLIENT — 6 tuples

ClientNo	FirstName	LastName	DOB	Phone	Address	LicenseNo
C001	Robert	Thomson	1982-03-14	0131-555-2001	Edinburgh	DL-100001
C002	Sarah	Wilson	1990-07-28	0141-555-2002	Glasgow	DL-100002
C003	David	Brown	1975-11-03	01224-555-2003	Aberdeen	DL-100003
C004	Laura	Mitchell	1998-05-19	01382-555-2004	Dundee	DL-100004
C005	Mark	Henderson	1985-09-30	01463-555-2005	Inverness	DL-100005
C006	Emma	Scott	1992-01-15	0131-555-2006	Edinburgh	DL-100006

3.5 HIRE_AGREEMENT — 7 tuples

HireNo	StartDate	EndDate	MilesBefore	MilesAfter	ClientNo	RegNumber
H001	2024-01-10	2024-01-15	12000	12350	C001	EH21 XYZ
H002	2024-02-05	2024-02-10	23000	23480	C002	GW19 ABC
H003	2024-03-01	2024-03-07	45000	45620	C003	AB18 DEF
H004	2024-04-12	2024-04-14	8000	8210	C004	DD20 GHI
H005	2024-05-20	2024-05-25	31000	31540	C001	EH22 JKL
H006	2024-06-01	2026-04-30	17500	NULL	C005	GW21 MNO
H007	2024-06-05	2026-04-30	52000	NULL	C006	AB20 PQR

H006 and H007 have NULL MileageAfter values, indicating the vehicles have not yet been returned. These are the active hire agreements targeted by Query 5.

4. Embedded SQL Queries

All five queries are implemented in `queries/queries.py` using Python's `sqlite3` library with parameterized `cursor.execute()` calls. Each query maps directly to a user transaction from the logical design report. Results are printed to `stdout` and captured by the GitHub Actions workflow for display in the repository README.

Query 1 — Vehicles at a Given Outlet

Lists all vehicles currently stocked at a given outlet along with their daily hire rates. Targets the Edinburgh outlet (O001). Joins `VEHICLE` to `OUTLET` on the `OutletNumber` foreign key.

```
SELECT v.RegistrationNumber, v.Make, v.Model,
       v.EngineSize, v.Capacity, v.DailyHireRate
FROM VEHICLE v
JOIN OUTLET o ON v.OutletNumber = o.OutletNumber
WHERE o.OutletNumber = 'O001'
```

Returns 2 rows: EH21 XYZ (Vauxhall Corsa, 35.00/day) and EH22 JKL (Volkswagen Golf, 50.00/day).

Query 2 — Hire History for a Specific Client

Retrieves the full hire history for a given client including all vehicles involved. Targets client C001 (Robert Thomson). Traverses `CLIENT` to `HIRE_AGREEMENT` to `VEHICLE` using two foreign key joins.

```
SELECT h.HireNumber, h.StartDate, h.EndDate,
       h.MileageBefore, h.MileageAfter,
       v.Make, v.Model, v.RegistrationNumber
FROM HIRE_AGREEMENT h
JOIN CLIENT c ON h.ClientNumber = c.ClientNumber
JOIN VEHICLE v ON h.RegistrationNumber = v.RegistrationNumber
WHERE c.ClientNumber = 'C001'
```

Returns 2 rows: H001 (Vauxhall Corsa, Jan 2024) and H005 (Volkswagen Golf, May 2024).

Query 3 — Staff at a Specific Outlet

Finds all staff members based at a given outlet with their job titles and salaries. Targets the Glasgow outlet (O002). Joins `STAFF` to `OUTLET` on the `OutletNumber` foreign key.

```

SELECT s.StaffNumber, s.FirstName, s.LastName,
       s.JobTitle, s.Salary
FROM STAFF s
JOIN OUTLET o ON s.OutletNumber = o.OutletNumber
WHERE o.OutletNumber = 'O002'

```

Returns 2 rows: S003 (Hamish Stewart, Branch Manager, 44000) and S004 (Morag Fraser, Sales Advisor, 27000).

Query 4 — Mileage Driven During a Hire Agreement

Calculates the total mileage driven during a specific hire agreement by subtracting MileageBefore from MileageAfter inline. Only executes when MileageAfter is not NULL. Targets hire H003 and retrieves client and vehicle details via foreign key joins.

```

SELECT h.HireNumber, h.MileageBefore, h.MileageAfter,
       (h.MileageAfter - h.MileageBefore) AS MileageDriven,
       c.FirstName, c.LastName, c.PhoneNumber,
       v.Make, v.Model, v.EngineSize
FROM HIRE_AGREEMENT h
JOIN CLIENT c ON h.ClientNumber = c.ClientNumber
JOIN VEHICLE v ON h.RegistrationNumber = v.RegistrationNumber
WHERE h.HireNumber = 'H003'
AND h.MileageAfter IS NOT NULL

```

Returns 1 row: David Brown, Nissan Qashqai, 620 miles driven (45000 to 45620).

Query 5 — Active Hire Agreements at a Given Outlet

Finds all currently active hire agreements for vehicles based at a given outlet, along with the client details for each. This is the most complex query in the implementation, joining four tables. Targets Aberdeen (O003). Active agreements are identified by a NULL MileageAfter and an EndDate on or after today.

```

SELECT h.HireNumber, h.StartDate, h.EndDate,
       v.RegistrationNumber, v.Make, v.Model,
       c.FirstName, c.LastName, c.PhoneNumber
FROM HIRE_AGREEMENT h
JOIN VEHICLE v ON h.RegistrationNumber = v.RegistrationNumber
JOIN OUTLET o ON v.OutletNumber = o.OutletNumber
JOIN CLIENT c ON h.ClientNumber = c.ClientNumber
WHERE o.OutletNumber = 'O003'
AND h.MileageAfter IS NULL
AND h.EndDate >= DATE('now')

```

Returns 1 row: H007, Emma Scott, Ford Kuga (AB20 PQR), active until 30 April 2026.

5. Database Contents

The following screenshots show the full contents of each relation, captured by running SELECT * queries against the live database after seed data insertion. The queries used to generate these screenshots are listed below.

```
SELECT * FROM OUTLET;  
SELECT * FROM VEHICLE;  
SELECT * FROM STAFF;  
SELECT * FROM CLIENT;  
SELECT * FROM HIRE_AGREEMENT;
```

5.1 OUTLET

```
1 SELECT * FROM OUTLET;
```

	OutletNumber	Address	PhoneNumber	FaxNumber
1	O001	12 High Street, Edinburgh, EH1 1TB	0131-555-0101	0131-555-0102
2	O002	45 Sauchiehall Street, Glasgow, G2 ...	0141-555-0201	0141-555-0202
3	O003	8 Union Street, Aberdeen, AB10 1DQ	01224-555-0301	01224-555-0302
4	O004	22 Princes Street, Dundee, DD1 4ER	01382-555-0401	01382-555-0402
5	O005	5 Castle Road, Inverness, IV2 3EA	01463-555-0501	01463-555-0502

```
Execution finished without errors.  
Result: 5 rows returned in 18ms  
At line 1:  
SELECT * FROM OUTLET;
```

5.2 VEHICLE

```
1 SELECT * FROM VEHICLE;
```

	RegistrationNumber	Model	Make	EngineSize	Capacity	CurrentMileage	DailyHireRate	OutletNumber
1	EH21 XYZ	Corsa	Vauxhall	1.2L	5	12000	35.0	0001
2	GW19 ABC	Focus	Ford	1.6L	5	23000	40.0	0002
3	AB18 DEF	Qashqai	Nissan	1.5L	5	45000	55.0	0003
4	DD20 GHI	Fiesta	Ford	1.0L	5	8000	30.0	0004
5	EH22 JKL	Golf	Volkswagen	2.0L	5	31000	50.0	0001
6	GW21 MNO	Astra	Vauxhall	1.4L	5	17500	38.0	0002
7	AB20 PQR	Kuga	Ford	2.0L	7	52000	60.0	0003

Execution finished without errors.
Result: 7 rows returned in 11ms
At line 1:
SELECT * FROM VEHICLE;

5.3 STAFF

```
1 SELECT * FROM STAFF;
```

	StaffNumber	FirstName	LastName	HomeAddress	HomePhoneNumber	DateOfBirth	Sex	DateJoined
1	S001	James	MacDonald	10 Elm Street, Edinburgh, EH3 5TN	0131-555-1001	1985-06-15	M	2010-03-01
2	S002	Fiona	Campbell	3 Oak Avenue, Edinburgh, EH4 2PL	0131-555-1002	1990-11-22	F	2015-07-01
3	S003	Hamish	Stewart	77 Birch Lane, Glasgow, G3 6QR	0141-555-1003	1978-04-30	M	2008-01-01
4	S004	Morag	Fraser	14 Pine Road, Glasgow, G4 9ST	0141-555-1004	1995-08-10	F	2019-09-01
5	S005	Callum	Ross	29 Cedar Close, Aberdeen, AB11 7UV	01224-555-1005	1988-02-17	M	2012-05-01
6	S006	Isla	Murray	6 Maple Drive, Aberdeen, AB12 3WX	01224-555-1006	1993-12-05	F	2018-04-01

Execution finished without errors.
Result: 6 rows returned in 8ms
At line 1:
SELECT * FROM STAFF;

5.4 CLIENT

```
1 SELECT * FROM CLIENT;
```

	ClientNumber	FirstName	LastName	HomeAddress	PhoneNumber	DateOfBirth	DrivingLicens
1	C001	Robert	Thomson	55 Baker Street, Edinburgh, EH6 4YZ	0131-555-2001	1982-03-14	DL-100001
2	C002	Sarah	Wilson	8 Victoria Road, Glasgow, G5 0AB	0141-555-2002	1990-07-28	DL-100002
3	C003	David	Brown	33 Albert Terrace, Aberdeen, AB13 ...	01224-555-2003	1975-11-03	DL-100003
4	C004	Laura	Mitchell	19 George Square, Dundee, DD2 1EF	01382-555-2004	1998-05-19	DL-100004
5	C005	Mark	Henderson	42 Queen Street, Inverness, IV3 8GH	01463-555-2005	1985-09-30	DL-100005
6	C006	Emma	Scott	7 King Avenue, Edinburgh, EH7 2IJ	0131-555-2006	1992-01-15	DL-100006

Execution finished without errors.
Result: 6 rows returned in 6ms
At line 1:
SELECT * FROM CLIENT;

5.5 HIRE_AGREEMENT

```
1 SELECT * FROM HIRE_AGREEMENT;
```

	HireNumber	StartDate	EndDate	MileageBefore	MileageAfter	ClientNumber	RegistrationNumber
1	H001	2024-01-10	2024-01-15	12000	12350	C001	EH21 XYZ
2	H002	2024-02-05	2024-02-10	23000	23480	C002	GW19 ABC
3	H003	2024-03-01	2024-03-07	45000	45620	C003	AB18 DEF
4	H004	2024-04-12	2024-04-14	8000	8210	C004	DD20 GHI
5	H005	2024-05-20	2024-05-25	31000	31540	C001	EH22 JKL
6	H006	2024-06-01	2026-04-30	17500	NULL	C005	GW21 MNO
7	H007	2024-06-05	2026-04-30	52000	NULL	C006	AB20 PQR

Execution finished without errors.
Result: 7 rows returned in 11ms
At line 1:
SELECT * FROM HIRE_AGREEMENT;

5.6 Query Results

```
$ python main.py

=====
Query 1 - Vehicles at Edinburgh (0001)
=====

SQL:

SELECT
    v.RegistrationNumber,
    v.Make,
    v.Model,
    v.EngineSize,
    v.Capacity,
    v.DailyHireRate
FROM VEHICLE v
JOIN OUTLET o ON v.OutletNumber = o.OutletNumber
WHERE o.OutletNumber = '0001'

Results:
RegistrationNumber | Make      | Model | EngineSize | Capacity | DailyHireRate
-----
EH21 XYZ          | Vauxhall  | Corsa | 1.2L       | 5        | 35.0
EH22 JKL          | Volkswagen | Golf  | 2.0L       | 5        | 50.0

=====
Query 2 - Hire history for Client C001 (Robert Thomson)
=====

SQL:

SELECT
    h.HireNumber,
    h.StartDate,
    h.EndDate,
    h.MileageBefore,
    h.MileageAfter,
    v.Make,
    v.Model,
    v.RegistrationNumber
FROM HIRE_AGREEMENT h
JOIN CLIENT c ON h.ClientNumber = c.ClientNumber
JOIN VEHICLE v ON h.RegistrationNumber = v.RegistrationNumber
WHERE c.ClientNumber = 'C001'

Results:
HireNumber | StartDate | EndDate | MileageBefore | MileageAfter | Make      | Model | RegistrationNumber
-----
H001       | 2024-01-10 | 2024-01-15 | 12000         | 12350        | Vauxhall  | Corsa | EH21 XYZ
```

```
=====
Query 3 - Staff at Glasgow (0002)
=====
```

SQL:

```
SELECT
    s.StaffNumber,
    s.FirstName,
    s.LastName,
    s.JobTitle,
    s.Salary
FROM STAFF s
JOIN OUTLET o ON s.OutletNumber = o.OutletNumber
WHERE o.OutletNumber = '0002'
```

Results:

StaffNumber	FirstName	LastName	JobTitle	Salary
S003	Hamish	Stewart	Branch Manager	44000.0
S004	Morag	Fraser	Sales Advisor	27000.0

```
=====
Query 4 - Mileage driven on Hire H003
=====
```

SQL:

```
SELECT
    h.HireNumber,
    h.MileageBefore,
    h.MileageAfter,
    (h.MileageAfter - h.MileageBefore) AS MileageDriven,
    c.FirstName,
    c.LastName,
    c.PhoneNumber,
    v.Make,
    v.Model,
    v.EngineSize
FROM HIRE_AGREEMENT h
JOIN CLIENT c ON h.ClientNumber = c.ClientNumber
JOIN VEHICLE v ON h.RegistrationNumber = v.RegistrationNumber
WHERE h.HireNumber = 'H003'
AND h.MileageAfter IS NOT NULL
```

Results:

HireNumber	MileageBefore	MileageAfter	MileageDriven	FirstName	LastName	PhoneNumber	Make	Model	EngineSize
H003	45000	45620	620	David	Brown	01224-555-2003	Nissan	Qashqai	1.5L

=====

Query 5 - Active hires at Aberdeen (0003)

=====

SQL:

```
SELECT
    h.HireNumber,
    h.StartDate,
    h.EndDate,
    v.RegistrationNumber,
    v.Make,
    v.Model,
    c.FirstName,
    c.LastName,
    c.PhoneNumber
FROM HIRE_AGREEMENT h
JOIN VEHICLE v ON h.RegistrationNumber = v.RegistrationNumber
JOIN OUTLET o ON v.OutletNumber = o.OutletNumber
JOIN CLIENT c ON h.ClientNumber = c.ClientNumber
WHERE o.OutletNumber = '0003'
AND h.MileageAfter IS NULL
AND h.EndDate >= DATE('now')
```

Results:

HireNumber	StartDate	EndDate	RegistrationNumber	Make	Model	FirstName	LastName	PhoneNumber
H007	2024-06-05	2026-04-30	AB20 PQR	Ford	Kuga	Emma	Scott	0131-555-2006

6. GitHub Repository and Automation

The complete implementation is publicly available at:

| <https://github.com/Elk-dev/DSC623part3>

6.1 GitHub Actions Workflow

A workflow defined in `.github/workflows/run_db.yml` triggers automatically on every push to the main branch. It sets up a clean Ubuntu environment, installs Python 3.12, runs the full script, captures the output, and injects the results between marker tags in `README.md` before committing and pushing the updated file back to the repository.

6.2 Live Results in README

The repository homepage displays live query results updated on every commit. Each query section shows the SQL followed by a formatted output table. This provides transparent, verifiable evidence of correct database operation without requiring the reader to run any code locally.

run-database

succeeded 2 weeks ago in 5s

- > ☒ Set up job
- > ☒ Checkout repository
- > ☒ Set up Python
- ▼ ☒ Run database and capture results

```
1 ▼ Run python src/main.py > query_output.txt
2   python src/main.py > query_output.txt
3   shell: /usr/bin/bash -e {0}
4   env:
5     pythonLocation: /opt/hostedtoolcache/Python/3.12.13/x64
6     PKG_CONFIG_PATH: /opt/hostedtoolcache/Python/3.12.13/x64/lib/pkgconfig
7     Python_ROOT_DIR: /opt/hostedtoolcache/Python/3.12.13/x64
8     Python2_ROOT_DIR: /opt/hostedtoolcache/Python/3.12.13/x64
9     Python3_ROOT_DIR: /opt/hostedtoolcache/Python/3.12.13/x64
10    LD_LIBRARY_PATH: /opt/hostedtoolcache/Python/3.12.13/x64/lib
```

- > ☒ Inject results into README
- > ☒ Commit updated README
- > ☒ Post Set up Python
- > ☒ Post Checkout repository
- > ☒ Complete job

7. Conclusion

The Reliable Rentals database has been fully implemented using Python and SQLite with embedded SQL. All five relations from the logical design have been created with their complete constraint sets, populated with representative seed data, and validated through five operational queries that map directly to the user transactions defined in the previous phase.

The GitHub Actions pipeline provides automated, repeatable execution with live results published to the repository README, giving any reader immediate access to the query output without needing to run the code themselves.